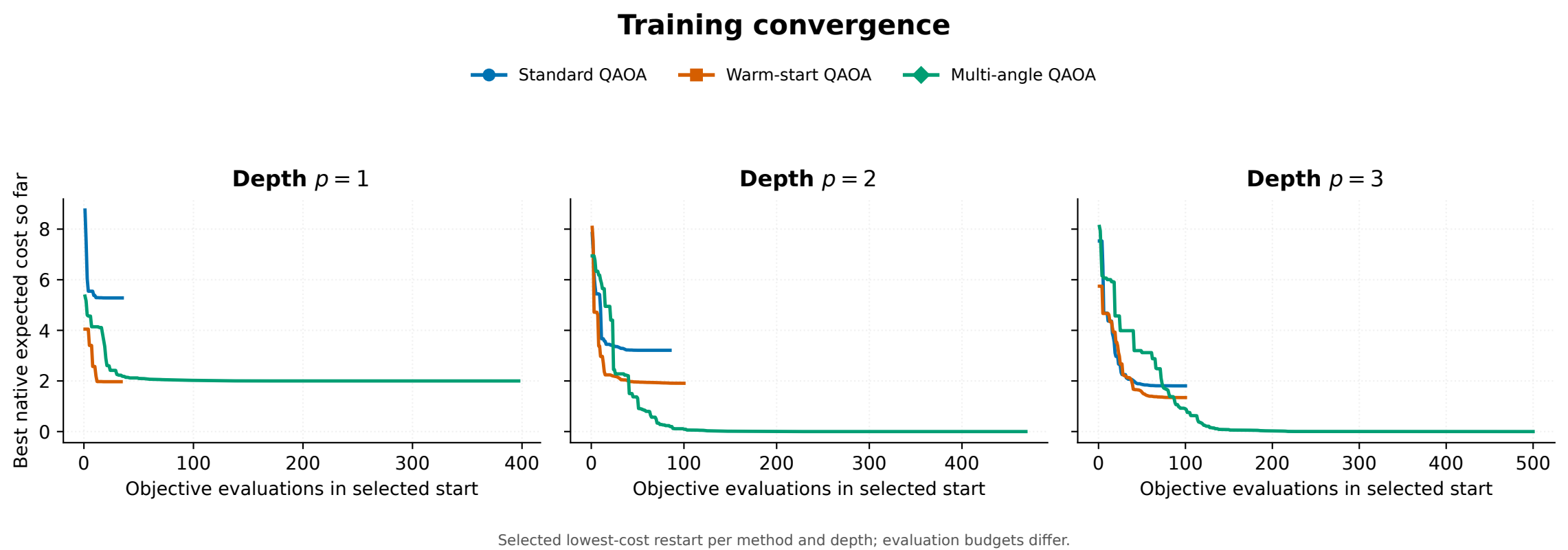

Training convergence
Standard QAOA
Warm-start QAOA
Multi-angle QAOA
Depth $p = 1$
Depth $p = 2$
Depth $p = 3$
Best native expected cost so far
0
2
4
6
8
0
100
200
300
400
500
Objective evaluations in selected start
Selected lowest-cost restart per method and depth; evaluation budgets differ.